AI Technology Stack Recommendation

6.1 Brief Description of Uploaded Materials

We uploaded our class diagram and system description featuring cross-platform driver recommendations, anonymous error reporting, TLS-secured operator authentication, background polling of public datasets, and administrative overrides.

6.2 Prompt & AI Response Transcript

Text Prompt Provided to AI:

> *"Based on the updated design materials for ParkChoiceSG (which include anonymous mobile driver features, high-frequency external API polling, and a secure authenticated Operator console for report management and manual overrides), recommend a technology stack appropriate for an entry-level software engineering team. We are familiar with Java, React, and modular backend architectures."*

AI Chat Response:

> ***Recommended Technology Stack:***
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> 1. ***Mobile Client**: **Flutter (Dart)** for cross-platform iOS and Android mobile app deployment with platform-adaptive navigation intents.*
> 2. ***Backend Application Server**: **Spring Boot (Java)** or **FastAPI (Python)** to manage REST endpoints, JWT-based operator authentication, and background async data polling.*
> 3. ***Caching & Queue Layer**: **Redis** for in-memory caching of high-frequency LTA DataMall lot readings and token-bucket rate limiting for user report submissions.*
> 4. ***Database Store**: **PostgreSQL** to maintain relational integrity between carparks, historical vacancy logs, operator audit trails, and user error reports.*
> 5. ***External Interfaces**: **OneMap SG API** (geocoding), **Google Maps Distance Matrix** (durations), and **LTA DataMall** (live availability).*

6.3 Written Opinion on Proposed Non-Language Technology (PostgreSQL)

Technology Selected: PostgreSQL (Relational Database Management System)

We feel that integrating **PostgreSQL** is good and suitable for ParkChoiceSG because our system requires structured relational data management for operator accounts, audit-logged carpark overrides, user-submitted error reports as well as static carpark metadata. For a student engineering/computing team with a high combined experience in using SQL Databases, we believe that PostgreSQL provides reliable data consistency, straightforward foreign-key relationships for report queues and low configuration difficulty within standard free-tier hosting platforms like Supabase.